

Beyond a Single Outcome: Evaluating Behavioral Learner Profiles in OULAD Against Academic Success, Engagement, and Course Completion

Batuhan C. Madenci

Intelligent Systems Engineering, Graduate Education Institute
Bursa Technical University
Bursa, Turkey
24434998030@ogr.btu.edu.tr

Abstract

Clustering is widely used to surface learner profiles in virtual learning environments, but the resulting profiles are typically validated against a single outcome—most often academic performance. We apply k -medoids (Partitioning Around Medoids) clustering to four behavioral features derived from the Open University Learning Analytics Dataset (OULAD; $N = 32,593$ student–module registrations) and evaluate the resulting profiles across three distinct outcome dimensions: academic success, engagement, and course completion/withdrawal. Four robust, well-separated profiles emerge—*disengaged non-starters*, *struggling partial participants*, *consistent completers*, and *highly engaged achievers*—differing significantly on every continuous outcome (Kruskal–Wallis, all $p < .001$). Critically, the profiles also discriminate course outcome, a variable never used in clustering (χ^2 test, Cramér’s $V = 0.51$), and they separate *withdrawal* from *failure* as distinct risk pathways. We further observe a diminishing-returns relationship between engagement and attainment that a single-outcome evaluation would obscure. The contribution is not a new algorithm but a more complete, multi-outcome evaluation framework for behavioral learner profiles.

Keywords

learning analytics, clustering, k -medoids, OULAD, student engagement, course completion, learner profiles

1 Introduction

Online learning environments now record learners’ interactions—clickstreams, assessment submissions, and access logs—at a scale unimaginable a decade ago. Yet this abundance does not translate automatically into insight: the central challenge is identifying which behavioral patterns are educationally meaningful, and for whom. Clustering offers a principled, data-driven response. By grouping students with similar behavioral profiles, it renders latent structure in trace data visible and interpretable, and has become a staple of learning-analytics practice on datasets such as the Open University Learning Analytics Dataset (OULAD) [5].

A recurring limitation, however, concerns how the resulting profiles are *evaluated*. Clustering studies on OULAD typically validate the discovered profiles against a single outcome—most often academic performance, operationalized as final grade or pass/fail status [6, 10]. Where engagement enters, it is generally an *input* that defines the clustering features rather than an outcome tested against cluster membership after the fact [1, 6]. Course completion and withdrawal are seldom evaluated as a dimension distinct from

performance, even though dropout risk carries instructional relevance independent of grades. This narrow evaluation limits the practical utility of the profiles: a structure that appears meaningful against performance may or may not be meaningful against engagement or retention.

This study addresses that limitation. We apply k -medoids clustering to four behavioral features from OULAD and evaluate the resulting profiles across three distinct outcome dimensions—academic success, engagement, and course completion/withdrawal—rather than against performance alone. The contribution is not a new algorithm but a more complete evaluation framework: we ask whether the same cluster structure is meaningful across multiple educationally relevant outcomes simultaneously, and we show that doing so surfaces distinctions—notably, withdrawal versus failure—that a single-outcome evaluation conflates. We address four research questions:

- RQ1** What distinct behavioral profiles emerge from students’ VLE interaction and assessment data?
- RQ2** Do the profiles differ in academic success (assessment scores)?
- RQ3** Do the profiles differ in engagement (VLE clicks and active days)?
- RQ4** Do the profiles differ in course completion and withdrawal?

2 Related Work

2.1 Clustering for Learner Profiling

Unsupervised clustering is a common approach for deriving learner typologies from behavioral trace data, partitioning students into interpretable groups without predefined labels. On OULAD and similar datasets, partitioning methods—particularly k -means—dominate, often paired with internal indices such as the silhouette width for selecting the number of clusters [1]. Dissimilarity-based alternatives such as k -medoids are comparatively underused despite their robustness to the heavy-tailed activity distributions typical of clickstream data [7]; we adopt k -medoids here for precisely this reason.

2.2 Single-Outcome Validation on OULAD

A consistent pattern across OULAD clustering work is the reliance on a single validation target. Zhang et al. [10] categorize e-learning behaviors in STEM modules and relate the categories to performance. Lima et al. [6] cluster students with k -means, EM, and

Table 1: Behavioral features and their associated constructs.

Feature	Source	Construct
total_clicks	studentVle	Engagement volume
active_days	studentVle	Engagement consistency
submission_rate	assessment tables	Assessment participation
mean_score	studentAssessment	Academic attainment

agglomerative methods and report a positive association between engagement and academic performance, relating the profiles to the final result as a single outcome. Alzahrani et al. [1] profile weekly engagement with k -means and validate the clusters against engagement labels derived from activity weighting, again collapsing evaluation onto one construct. In each case engagement serves chiefly as a clustering *input* or as the sole validation target, and course completion is not assessed as a dimension distinct from performance. Consequently, whether a single behavioral profile structure is simultaneously meaningful for attainment, engagement, and retention—and whether withdrawal and failure form separable pathways—remains open. The present study targets exactly this gap.

3 Methods

3.1 Dataset

We use the Open University Learning Analytics Dataset (OULAD) [5], released under a CC-BY 4.0 license. OULAD contains anonymized records for 32,593 students across 22 module presentations, including roughly 10.6 million daily Virtual Learning Environment (VLE) interaction entries. Four tables drive the analysis: studentInfo (final course outcome and demographics), studentVle (daily clickstream logs), studentAssessment (assessment submissions and scores), and assessments (assessment metadata).

The **unit of analysis** is a single *student–module–presentation registration*: a student enrolled in two modules contributes two independent rows, since behavior may differ across courses.

3.2 Feature Engineering

From the raw tables we derive the four behavioral features in Table 1, spanning two constructs anticipated by their pairwise correlations—behavioral engagement and assessment attainment.

Exclusion of exams. OULAD assessments fall into three types (TMA, CMA, Exam). Final-exam scores are systematically missing for most students; including exams in the submission_rate denominator would deflate the rate artificially and introduce a structural bias. We therefore compute assessment features over TMA and CMA only. No information is lost, because the final-exam outcome is captured independently through final_result (tested in RQ4), keeping the clustering inputs and the RQ4 outcome non-redundant. The submission-rate denominator is computed *per module–presentation* (the count of TMA/CMA assessments actually defined for that course), yielding a rate bounded in $[0, 1]$.

Table 2: Distribution of the final course outcome (final_result).

Outcome	n	%
Pass	12,361	37.9
Withdrawn	10,156	31.2
Fail	7,052	21.6
Distinction	3,024	9.3
Total	32,593	100.0

3.3 Missing-Value Treatment

Missingness in OULAD is not random. Among withdrawn students, 29.4% have no VLE record and 54.0% have no assessment record, whereas the corresponding rates are 0% for both passing and distinction students. Listwise deletion would therefore remove precisely the disengaged learners that RQ4 is designed to study, biasing the very outcome under analysis. We instead treat missing values as *informative zeros*: no VLE activity is encoded as zero clicks and zero active days, and non-submission as a submission rate and demonstrated-attainment score of zero. A student who submitted one assessment and scored zero remains distinguishable from a non-submitter, because the two differ on submission_rate. (The raw score field was stored as a string; it was coerced to numeric, with non-numeric tokens treated as missing.)

3.4 Outlier Handling and Standardization

k -medoids represents each cluster by an actual observation rather than a mean, making it inherently robust to the heavy right tail of the clickstream features; aggressive outlier removal would risk discarding a legitimate high-engagement profile. We therefore remove no students. All four features are standardized via z -score normalization so that no feature dominates the Euclidean dissimilarity through scale alone. Engagement features are moderately intercorrelated ($r = 0.84$ for clicks vs. active days; $r = 0.76$ for submission rate vs. score), which is acceptable for clustering but is noted as a limitation (Section 5).

3.5 Descriptive Statistics

Table 2 summarizes the distribution of the final course outcome across the 32,593 registrations. Passing students form the plurality (37.9%), but withdrawals (31.2%) and failures (21.6%) together account for more than half the cohort, underscoring why retention is a substantively important outcome and not a rare edge case. Distinctions are comparatively scarce (9.3%).

The four behavioral features are all positively intercorrelated (Table 3), with the strongest associations between the two engagement measures ($r = 0.84$) and between assessment participation and attainment ($r = 0.76$). This block structure indicates that the features load onto two broad latent dimensions—behavioral engagement and assessment attainment—rather than four independent ones, a property that informs both the expected cluster geometry (Section 4) and the limitations (Section 5). The engagement features are also strongly right-skewed: median total_clicks (739.5) sits far below the mean (1,355.0), reflecting a minority of very high-activity

Table 3: Pearson correlations among the standardized behavioral features.

	Clicks	Days	Sub. rate	Score
Clicks	1.00	0.84	0.51	0.44
Days	0.84	1.00	0.68	0.58
Sub. rate	0.51	0.68	1.00	0.76
Score	0.44	0.58	0.76	1.00

students—precisely the regime in which k -medoids’ robustness is advantageous.

3.6 Clustering (RQ1)

We cluster students with k -medoids (Partitioning Around Medoids) [3] on the standardized features using Euclidean dissimilarity, following the dissimilarity-based framework of Murphy et al. [7]. The implementation uses the FasterPAM algorithm [9].

The number of clusters is selected by the Average Silhouette Width (ASW) [8], which is dissimilarity-based and algorithm-agnostic; we deliberately avoid the Elbow method (tied to k -means inertia and outlier-sensitive) and information criteria such as BIC/AIC (which assume a probabilistic mixture model mismatched to a dissimilarity-based method). Because a full pairwise dissimilarity matrix over all 32,593 students is computationally prohibitive (~ 8 GB), ASW is evaluated on a representative random sample ($n = 8,000$); the medoids are then fit at the chosen k and *all* students are assigned to their nearest medoid, so no student is dropped from the outcome analysis.

3.7 Outcome Comparisons (RQ2–RQ4)

Each research question is tested with a method matched to the outcome’s measurement level. The two continuous outcomes (RQ2: `mean_score`; RQ3: `total_clicks`, `active_days`) use the Kruskal–Wallis H test [4], a non-parametric alternative to one-way ANOVA that does not assume normality; effect size is reported as ϵ^2 . Significant omnibus tests are followed by Dunn’s post-hoc test with Bonferroni correction [2]. The categorical outcome (RQ4: `final_result`) uses the χ^2 test of independence with Cramér’s V as effect size.

We emphasize interpretation: `mean_score` (RQ2) and the engagement features (RQ3) were themselves clustering inputs, so differences there *characterize and confirm* the profile structure rather than validate it independently. The genuinely external validation is RQ4, since `final_result` was never used in clustering. Given $N \approx 32,600$, where even trivial differences reach significance, we foreground effect sizes over p -values throughout.

4 Results

4.1 RQ1: Behavioral Profiles

ASW was marginally highest at $k = 2$ (0.496) but statistically indistinguishable at $k = 4$ (0.496; $\Delta \approx 0.0003$), declining thereafter (Figure 1). A two-cluster solution collapses to a single engaged/disengaged dichotomy, too coarse to support RQ3 and RQ4; we therefore select $k = 4$, which preserves silhouette quality while providing the profile resolution the research questions require.

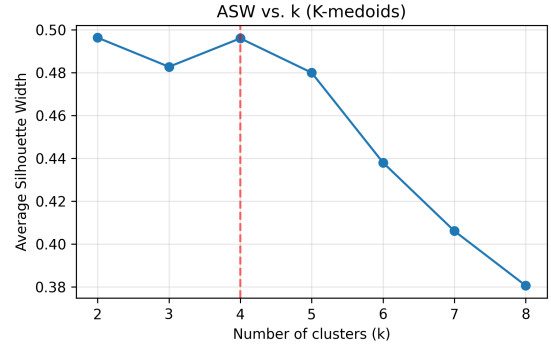


Figure 1: Average Silhouette Width (ASW) by number of clusters k . The near-tie between $k = 2$ and $k = 4$ motivates selecting $k = 4$ on interpretability grounds; ASW degrades for $k \geq 5$.

Table 4: Cluster behavioral profiles (raw feature means).

Profile	n	Clicks	Days	Sub. rate	Score
Disengaged non-starters	7,302	61.3	3.8	0.01	1.2
Struggling partial part.	6,779	456.0	24.5	0.29	67.7
Consistent completers	11,802	937.9	56.3	0.93	74.1
Highly engaged achievers	6,710	3,725.4	141.5	0.97	81.0

The four clusters (Table 4, Figure 2) correspond to substantively distinct learner types structured along the two anticipated axes:

- **Disengaged non-starters** ($n = 7,302$): minimal activity, near-zero submission, negligible scores—predominantly withdrawals and no-shows.
- **Consistent completers** ($n = 11,802$, largest): below-average browsing but high submission rate and solid scores—efficient, task-focused students.
- **Struggling partial participants** ($n = 6,779$): low engagement and low submission, yet moderate scores when they submit—a follow-through gap rather than an ability gap.
- **Highly engaged achievers** ($n = 6,710$): top values on every feature—intensive activity paired with high attainment.

Notably, completers and achievers *dissociate engagement from achievement*: both reach high attainment, but via opposite engagement levels. This is the kind of structure a single-outcome evaluation would obscure.

4.2 RQ2: Academic Success

Profiles differ significantly in mean assessment score (Kruskal–Wallis $H = 18,931.2$, $p < .001$, $\epsilon^2 = 0.581$). As a clustering input, this confirms separation along the attainment axis (Table 5).

4.3 RQ3: Engagement

Profiles differ significantly on both engagement measures: total clicks ($H = 25,121.1$, $p < .001$, $\epsilon^2 = 0.771$) and active days ($H = 26,603.8$, $p < .001$, $\epsilon^2 = 0.816$). Dunn post-hoc comparisons are

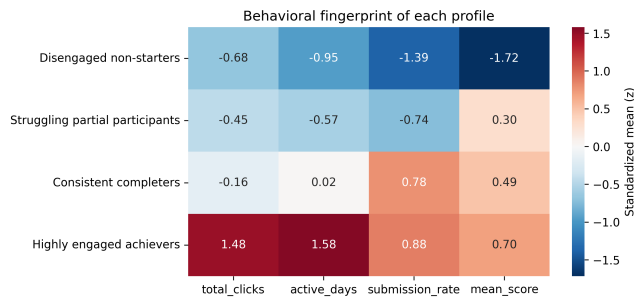


Figure 2: Standardized behavioral fingerprint of each profile (z-scores). High attainment is reached both through intensive activity (achievers) and through lean, targeted participation (completers).

Table 5: Inferential results by research question. KW: Kruskal–Wallis; effect sizes are ϵ^2 (KW) and Cramér’s V (χ^2).

RQ	Outcome	Statistic	p	Effect
RQ2	Mean score	$H = 18,931.2$	$< .001$	0.581
RQ3	Total clicks	$H = 25,121.1$	$< .001$	0.771
RQ3	Active days	$H = 26,603.8$	$< .001$	0.816
RQ4	Course outcome	$\chi^2 = 25,277.5$	$< .001$	0.508

significant for all six cluster pairs on every continuous outcome ($p < .001$), indicating that no two profiles overlap.

4.4 RQ4: Course Completion and Withdrawal (External Validation)

The behavioral profiles strongly discriminate final course outcome—a variable never used in clustering ($\chi^2(9) = 25,277.5$, $p < .001$, Cramér’s $V = 0.508$, a large effect). The outcome composition (Figure 3) shows a monotonic shift across the profile axis: *disengaged non-starters* are 80.0% withdrawn and 0.1% pass; *highly engaged achievers* are 91.7% successful (67.2% pass, 24.5% distinction) and only 1.9% withdrawn; *consistent completers* reach the highest pure-pass rate (65.2%).

Crucially, withdrawal and failure separate as distinct pathways: *struggling partial participants* split almost evenly between failing (45.4%) and withdrawing (52.0%), whereas *non-starters* overwhelmingly withdraw without failing. A single academic-performance outcome would conflate these two risk types; evaluating completion as a separate dimension reveals them. This is the central, externally-validated result of the study.

4.5 Engagement–Attainment Returns

Although completers and achievers both score highly, the relationship is one of diminishing returns: completers reach a mean score of 74.1 ($\approx 91\%$ of the achievers’ 81.0) with only 937.9 clicks ($\approx 25\%$ of the achievers’ 3,725.4). High attainment is thus reachable through lean, targeted participation, not only intensive activity—a pattern obscured when performance is the sole evaluation lens.

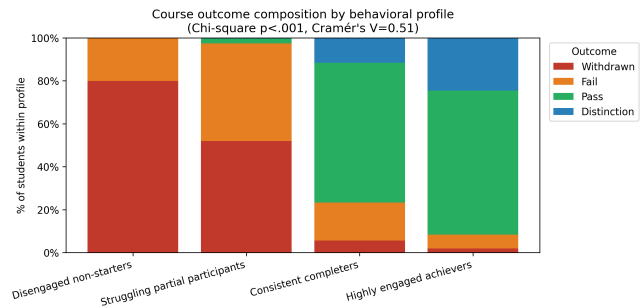


Figure 3: Course-outcome composition within each behavioral profile ($\chi^2 p < .001$, Cramér’s $V = 0.51$). Withdrawal (red) recedes and pass/distinction (green/blue) rise across the profile axis.

5 Discussion

5.1 Profiles Are Meaningful Across, Not Just Within, a Single Outcome

The four profiles differ significantly on every outcome we tested, but the informative comparisons are the external ones. Because mean_score and the engagement features were clustering inputs, their significance in RQ2 and RQ3 confirms rather than discovers structure. The substantive finding is RQ4: profiles built entirely from behavior also discriminate *course outcome*—a variable never shown to the clustering algorithm—with a large effect (Cramér’s $V = 0.51$). This is the evidence the proposal’s framing called for: the same cluster structure is meaningful across academic success, engagement, and completion simultaneously, rather than against performance alone.

5.2 Withdrawal and Failure Are Distinct Pathways

The most practically consequential result is that withdrawal and failure separate across profiles. *Disengaged non-starters* overwhelmingly withdraw (80.0%) without failing, while *struggling partial participants* split almost evenly between failing (45.4%) and withdrawing (52.0%). A single performance outcome—or a binary pass/fail target, as used in much prior OULAD clustering work [6, 10]—would collapse these two groups into one “unsuccessful” category and obscure that they need different interventions. Non-starters never meaningfully engage and call for early, pre-enrolment or first-week outreach; partial participants do engage and submit some work, so they are candidates for mid-course support targeting follow-through rather than re-recruitment. Treating completion as an outcome distinct from performance is what makes this actionable distinction visible.

5.3 Engagement Is Not Synonymous with Attainment

The dissociation between *consistent completers* and *highly engaged achievers* cautions against equating clickstream volume with learning. Completers reach roughly 91% of the achievers’ mean score with about a quarter of the clicks, indicating diminishing returns to

raw activity. Engagement dashboards that flag low-click students as at-risk would misclassify the efficient completers—the largest profile in our data—as needing intervention. This echoes the concern that behavioral volume is a noisy proxy for the construct of engagement [1] and supports evaluating attainment and activity as separate dimensions.

5.4 Limitations

Several limitations bound these claims. First, the clustering features are not orthogonal (engagement features correlate at $r = 0.84$), so the four profiles largely array along two latent axes rather than four independent ones; the substantive profiles remain interpretable, but the feature space is lower-dimensional than its nominal size. Second, RQ4 is only *partly* independent of the clustering inputs: course outcome correlates naturally with assessment score and submission rate, so withdrawal—which no feature measures directly—is the cleanest externally validated signal, while the pass/distinction contrast is less independent. Third, for scalability we selected k on a sample and assigned the full cohort by nearest medoid rather than running PAM over all 32,593 students; while standard, this is an approximation to a full-sample partition. Fourth, registrations are pooled across 22 module presentations of differing length and assessment structure; standardizing within module-presentation, or modelling module as a covariate, could refine the profiles. Fifth, the analysis is cross-sectional—features aggregate the whole presentation—so it cannot speak to *when* disengagement occurs, which a temporal or sequence-based extension could address. Sixth, the clustering features were limited to four aggregate behavioral indicators; richer feature sets—activity type breakdowns, temporal regularity, or forum participation—might reveal finer profiles, at the cost of the interpretability the present minimal set affords. Finally, the profiles are derived from a single dataset and institution; their generality to other platforms and populations is untested.

5.5 Future Work

Several extensions follow directly from these limitations. A temporal or sequence-based analysis—for example, hidden Markov models or sequence clustering over weekly activity—could locate *when* the disengaged non-starters and struggling participants diverge from successful learners, turning the static profiles into an early-warning signal usable before withdrawal occurs. Modelling module-presentation as a covariate, or clustering within standardized module-presentation strata, would separate genuinely cross-course behavioral types from artifacts of differing course structure. The multi-outcome evaluation framework itself is dataset-agnostic and should be replicated on other learning-analytics datasets to test whether the withdrawal-versus-failure dissociation generalizes. Finally, the practical value of the profiles—particularly the distinction between non-starters needing early outreach and partial participants needing follow-through support—should be assessed in an intervention study rather than inferred from cross-sectional association alone.

6 Conclusion

We applied k -medoids clustering to four behavioral features from OULAD and, rather than validating the resulting profiles against a single outcome, evaluated them across academic success, engagement, and course completion. Four robust, fully separated profiles emerged, and they discriminated course outcome—a variable never used in clustering—with a large effect, separating withdrawal from failure as distinct risk pathways and revealing diminishing returns of engagement to attainment. The contribution is methodological: a multi-outcome evaluation framework that surfaces structure a single-outcome validation would hide. Future work should pursue temporal and sequence-based extensions to locate *when* disengagement emerges, model module-presentation heterogeneity explicitly, and test whether the profiles transfer to other learning-analytics datasets.

Acknowledgments

The author used the AI assistant Claude (Anthropic) for translating the Chapter 8 analysis pipeline from R to Python, for debugging, and for drafting and editing assistance. All analytical decisions, code, and interpretations were reviewed and are the author's own responsibility, in line with the course policy on the disclosure of AI tools.

References

- [1] Nisreen Alzahrani, Maram Meccawy, Halima Samra, and Hassan A. El-Sabagh. 2025. Identifying weekly student engagement patterns in e-learning via k -means clustering and label-based validation. *Electronics* 14, 15 (2025), 3018. doi:10.3390/electronics14153018
- [2] Olive Jean Dunn. 1964. Multiple comparisons using rank sums. *Technometrics* 6, 3 (1964), 241–252.
- [3] Leonard Kaufman and Peter J. Rousseeuw. 1990. *Finding Groups in Data: An Introduction to Cluster Analysis*. John Wiley & Sons.
- [4] William H. Kruskal and W. Allen Wallis. 1952. Use of ranks in one-criterion variance analysis. *J. Amer. Statist. Assoc.* 47, 260 (1952), 583–621.
- [5] Jakub Kuzilek, Martin Hlosta, and Zdenek Zdrahal. 2017. Open University Learning Analytics dataset. *Scientific Data* 4 (2017), 170171. doi:10.1038/sdata.2017.171
- [6] Geycy D. O. Lima, Juliette A. R. Costa, Rafael D. Araújo, and Fabiano A. Dorça. 2023. Exploring the relationship between students' engagement and self-regulated learning: A case study using OULAD dataset and machine learning techniques. In *Anais do XXXIV Simpósio Brasileiro de Informática na Educação (SBIE)*. 1154–1165. doi:10.5753/sbie.2023.234344
- [7] Keefe Murphy, Sonsoles López-Pernas, and Mohammed Saqr. 2024. Dissimilarity-based cluster analysis of educational data: A comparative tutorial using R. In *Learning Analytics Methods and Tutorials*, Mohammed Saqr and Sonsoles López-Pernas (Eds.). Springer. doi:10.1007/978-3-031-54464-4_8
- [8] Peter J. Rousseeuw. 1987. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *J. Comput. Appl. Math.* 20 (1987), 53–65.
- [9] Erich Schubert and Peter J. Rousseeuw. 2021. Fast and eager k -medoids clustering: O(k) runtime improvement of the PAM, CLARA, and CLARANS algorithms. *Information Systems* 101 (2021), 101804. doi:10.1016/j.is.2021.101804
- [10] Jie Zhang, Feng Qiu, Wei Wu, Jing Wang, Rui Li, Min Guan, and Jin Huang. 2023. E-learning behavior categories and influencing factors of STEM courses: A case study of the OULAD. *Sustainability* 15, 10 (2023), 8235. doi:10.3390/su15108235